Problem 10 – Humanize (original statement)

1 Statement and strategy

Throughout, H denotes the *hyperbolic plane lattice*, that is the unique even unimodular lattice of signature $(1,1)$; we write it as

$$H \;=\; \mathbb{Z}\sigma \oplus \mathbb{Z}f, \qquad \sigma^2 = -2, \quad f^2 = 0, \quad \sigma \cdot f = 1,$$

and note the more usual form $H \simeq U$, the even unimodular hyperbolic plane (basis $e_1 = \sigma + f$, $e_2 = f$ with $e_1^2 = e_2^2 = 0$, $e_1 \cdot e_2 = 1$). In the moduli theory of Alexeev–Brunyate–Engel, H is the polarisation lattice of elliptic K3 surfaces with a section, so H-polarised K3 surfaces form an 18-dimensional family; the phrase "18+2" refers to this dimension 18 of moduli on the degeneration side and the rank 2 of the polarisation. We let E_8 denote the negative definite E_8 root lattice and

$$N \;\coloneqq\; H \oplus E_8 \oplus E_8,$$

an even unimodular lattice of signature $(1,17)$. Finally

$$\Lambda_{\mathrm{K3}} \;\coloneqq\; U^{\oplus 3} \oplus E_8^{\oplus 2}$$

denotes the K3 lattice (even, unimodular, signature $(3,19)$; we use the convention, in force throughout, that E_8 is negative definite, so in more standard notation this is $U^{\oplus 3} \oplus E_8(-1)^{\oplus 2}$).

The statement to be proved is the following.

Theorem 1.1. *Let $\mathcal{X} \to \Delta$ be an H-polarised Tyurin degeneration of K3 surfaces which is very general in the sense that $0 \in \Delta$ does not map into any Noether–Lefschetz locus of the Alexeev–Brunyate–Engel compactification of the moduli of H-polarised K3 surfaces. Let Y be a very general $H \oplus E_8 \oplus E_8$-quasipolarised K3 surface. Then there exists an elliptic fibration $\pi\colon Y \to \mathbb{P}^1$ and an allowable loop $\gamma \subset \mathbb{P}^1$, in the sense of* [GT, Definition 4.11]*, such that the $H \oplus E_8 \oplus E_8$-quasipolarisation is compatible with π and γ in the sense of* [GT, Definition 4.21]*, and such that $\mathcal{X} \to \Delta$ and $\pi\colon Y \to \mathbb{P}^1$ are a mirror pair in the sense of* [GT, Definition 5.1].

This is the case

$$(L, N) \;=\; \big(H,\; H \oplus E_8 \oplus E_8\big)$$

of the mirror duality between Tyurin degenerations of lattice-polarised K3 surfaces and elliptic K3 fibrations with an allowable loop developed by Giovenzana–Thompson. Our proof makes the mirror construction completely explicit in this special case. The argument has the following steps.

(1) *Lattice duality.* We compute the Dolgachev–Nikulin dual of N: the mirror lattice of N is H itself (Section 2). This is a purely lattice-theoretic verification of the duality condition of [GT, Definition 5.1(1)].

(2) *The fibration.* We prove that the very general N-quasipolarised K3 surface Y carries a unique elliptic fibration $\pi\colon Y \to \mathbb{P}^1$ with section, that its Néron–Severi lattice and its lattice of essential sections satisfy $\mathrm{NS}(Y) = N$ and $W(Y) = 0$, and that its singular fibres are exactly

$$II^* \;+\; II^* \;+\; 4\,I_1$$

(Section 3). The two E_8 summands of the quasipolarisation are realised as the root lattices of the two II^*-fibres.

(3) *The loop.* We show that a simple loop γ enclosing exactly the four I_1-points is allowable and computes a dual complex $\Gamma(Y,\gamma) \simeq S^2$, coinciding with the dual complex of the central fibre of $\mathcal{X} \to \Delta$, which we show is homeomorphic to S^2 as well (Sections 4 and 5).

(4) *Monodromy and weight filtrations.* We compute the filtrations $E_m \subset E'_m$ of [GT, Definition 5.1(4)] on both sides and exhibit an isometry between them respecting the full filtrations (Sections 4 and 5): on both sides E_m has rank 18 with W_2 the line generated respectively by the fibre class f and by the class of the double curve, and E'_m/E_m is a rank 2 lattice isomorphic to $U(-1)$.

(5) *Compatibility.* We exhibit the semiample class $h = \sigma + 2f$ in the H-summand of N and verify the compatibility conditions of [GT, Definition 4.21] for (π,γ) (Section 5).

(6) *Conclusion via the mirror theorem.* With all hypotheses verified, the mirror-pair statement follows from [GT, Theorem C], which we state precisely as Theorem 5.2 with all its hypotheses checked (Section 5).

External results used. We use the following classical or quoted results, each stated with its hypotheses where it is invoked: (a) Niemeier's classification of even unimodular lattices of signature $(2,2)$ and Nikulin's embedding theory; (b) the global Torelli theorem and the surjectivity of the period map for K3 surfaces, and the consequence that a very general lattice-polarised K3 surface has Néron–Severi lattice exactly the polarisation lattice; (c) Saint-Donat's analysis of linear systems on K3 surfaces; (d) Kodaira's classification of singular fibres and the Shioda–Tate formula; (e) the Kulikov–Persson–Pinkham description of Type III degenerations of K3 surfaces and Tyurin's comb models; (f) [GT, Definitions 4.11, 4.21, 5.1 and Theorem C], quoted precisely in the text. Everything else is proved.

2 The lattice duality

Proposition 2.1 (Dolgachev–Nikulin self-duality of H, cf. [GT, Definition 5.1(1)])**.** *There exist primitive embeddings*

$$N = H \oplus E_8^{\oplus 2} \hookrightarrow \Lambda_{\mathrm{K3}}, \qquad H \hookrightarrow \Lambda_{\mathrm{K3}}$$

with orthogonal complements

$$N^{\perp}_{\Lambda_{\mathrm{K3}}} \;\simeq\; U \oplus H, \qquad H^{\perp}_{\Lambda_{\mathrm{K3}}} \;\simeq\; U \oplus N.$$

In particular N and H are Dolgachev–Nikulin dual lattices, and the mirror lattice of the N-quasipolarised family is $\bar{N} = H$ (of rank 2), while the mirror lattice of the H-polarised family is $\bar{H} = N$ (of rank 18).

Proof. Embed $E_8 \oplus E_8$ identically into the $E_8 \oplus E_8$ summand of $\Lambda_{\mathrm{K3}} = U^{\oplus 3} \oplus E_8^{\oplus 2}$. It remains to exhibit a primitive embedding $H \hookrightarrow U^{\oplus 3}$ with $H^{\perp}_{U^{\oplus 3}} \simeq U \oplus H$. Since

$$U^{\oplus 2} \oplus U^{\oplus 2} \;=\; U^{\oplus 4}$$

has signature $(4,4)$, consider first the gluing construction: the even unimodular lattice $U^{\oplus 2}$ of signature $(2,2)$ contains the finite index sublattice $H \oplus H$, because $H \simeq U$ (both are even unimodular of signature $(1,1)$, and Niemeier's classification in signature $(1,1)$ gives uniqueness of the even unimodular lattice). Embed one copy $H_A = U$ into the first factor of $U \oplus U$, embedded diagonally as the U summand; then $H_A^{\perp} \subset U^{\oplus 3}$ contains the remaining $U^{\oplus 2}$. More precisely, write $U^{\oplus 3} = U_A \oplus U_B \oplus U_C$ and take $H_A \coloneqq U_A$. Then $H_A^{\perp} = U_B \oplus U_C \simeq U^{\oplus 2} \simeq U \oplus H$. For the second embedding take the same $H_A \subset \Lambda_{\mathrm{K3}}$; then

$$H_A^{\perp} \;=\; U_B \oplus U_C \oplus E_8^{\oplus 2} \;=\; U^{\oplus 2} \oplus E_8^{\oplus 2} \;=\; U \oplus \big(U \oplus E_8^{\oplus 2}\big) \;=\; U \oplus N,$$

because $U \simeq H$ implies $N = H \oplus E_8^{\oplus 2} = U \oplus E_8 \oplus E_8$. Primitivity holds because H (equivalently U) and E_8 are unimodular: a unimodular sublattice of a unimodular lattice is automatically a primitive orthogonal summand. All embeddings are unique up to isometry by Nikulin's refinement of Niemeier's theorem (the lattices involved are indefinite, unimodular, and satisfy his existence criterion, so the primitive embeddings exist and are unique up to the orthogonal group). □

Remark 2.2. *For an N-polarised K3 surface Y with $\mathrm{NS}(Y) = N$, the lattice of essential sections (in the sense of Harder–Thompson) of any elliptic fibration with section whose fibre lies in the H-summand satisfies $W(Y) = \big(U \oplus E_8^{\oplus 2}\big)^{\perp}_{\mathrm{NS}(Y)} = 0$, because $N = U \oplus E_8^{\oplus 2}$ is exactly the Néron–Severi lattice and $U \oplus E_8^{\oplus 2}$ has full rank 18. Thus the fibration has no deformation space: it is unique, as we verify directly in Section 3. The corresponding period domains are $\mathcal{D}_N$ (2-dimensional, an orthogonal Shimura variety for $N^{\perp} = U \oplus H$) and $\mathcal{D}_H$ (18-dimensional); these are Looijenga mirror period domains, and the mirror map between them is the "mirror correspondence" of [GT, (1.1)].*

3 The mirror fibration

In this section we prove:

Theorem 3.1. *Let Y be a very general $H \oplus E_8 \oplus E_8$-quasipolarised K3 surface. Then:*

(a) $\mathrm{NS}(Y) \simeq N = H \oplus E_8 \oplus E_8$*, with the H-summand generated by the class σ of a section and the class f of a smooth fibre of a genus-one fibration $\pi\colon Y \to \mathbb{P}^1$, and the E_8-summands generated by exceptional curves;*

(b) π has a section, $W(Y) = 0$ (so it is the unique such fibration), and its singular fibres are exactly

$$\pi^{-1}(0) \simeq II^*, \qquad \pi^{-1}(\infty) \simeq II^*, \qquad \pi^{-1}(p_i) \simeq I_1 \ (i = 1, \dots, 4),$$

for four distinct points $p_1, \dots, p_4 \in \mathbb{P}^1 \setminus \{0, \infty\}$; the roots of the two E_8-summands are precisely the components of the two II^*-fibres not meeting the section;

(c) the class $h \coloneqq \sigma + 2f \in H \subset N$ is big and nef, vanishes exactly on the two E_8-root systems, and defines a birational contraction of the two $\tilde{E}_8$-configurations; it is a semiample (quasi-ample) N-quasipolarisation;

(d) conversely the pair (Y, π) is realised by a Weierstrass model, and the very general such Y has the stated properties.

3.1 Very general lattice polarisation

By the surjectivity of the period map and the global Torelli theorem for lattice-polarised K3 surfaces, a very general N-quasipolarised K3 surface Y exists and satisfies $\mathrm{NS}(Y) = N$: indeed the period domain $\mathcal{D}_N$ for $N^\perp = U \oplus H$ (signature $(2,2)$) is a Shimura variety of orthogonal type, its Noether–Lefschetz loci are a countable union of proper subvarieties, and for periods outside all of them the Picard group equals N; one may also ensure that a chosen chamber of the positive cone of $N \otimes \mathbb{R}$ is the ample cone, so that the class f (see below) is nef. Concretely we can and do fix the embedding $N \hookrightarrow \Lambda_{\mathrm{K3}}$ of Proposition 2.1.

3.2 Existence and fibre type

Write the H-summand of N as $\mathbb{Z}\sigma \oplus \mathbb{Z}f$ with $\sigma^2 = -2$, $f^2 = 0$, $\sigma \cdot f = 1$.

Lemma 3.2. *The complete linear system $|f|$ is base-point-free and defines a genus-one fibration $\pi\colon Y \to \mathbb{P}^1$ with section class σ (after replacing σ by $\sigma + kf$ if necessary) and exactly the singular fibres stated in Theorem 3.1(b).*

Proof. Since $\mathrm{NS}(Y) = N$, the nef cone of Y is the closure of a chamber of $\{x \in N \otimes \mathbb{R} : x^2 > 0\}$ cut out by the roots of N. The roots of N are exactly the $240 + 240$ roots of the two E_8-summands; in particular f is a non-divisible isotropic vector with $f \cdot r = 0$ for every root r, so f is nef (it lies in the closure of every chamber that contains a class pairing positively with all effective (-2)-curves, e.g. the chamber of $\sigma + 3f$). By Saint-Donat's theorem, for a nef isotropic non-divisible class f on a K3 surface with $|f| \neq \emptyset$, the linear system $|f|$ is base-point-free and defines a genus-one fibration $\pi\colon Y \to \mathbb{P}^1$; positivity of the chamber containing $\sigma + 3f$ gives effectiveness of f (Riemann–Roch: $h^0(\mathcal{O}_Y(f)) \geq 2 + \frac{1}{2}f^2 = 2$).

Section. Since $\sigma^2 = -2$, Riemann–Roch gives $h^0(\mathcal{O}_Y(\sigma)) + h^0(\mathcal{O}_Y(-\sigma)) \geq 2 + \frac{1}{2}\sigma^2 = 1$, so σ or $-\sigma$ is effective. Applying if necessary a product of reflections in roots of the two E_8-summands (which fix f and h, hence preserve nefness of both), we may assume σ itself effective. Then σ is represented by an effective divisor C of arithmetic genus $p_a(C) = 1 + \frac{1}{2}\sigma^2 = 0$; since $C \cdot f = 1$ and f is nef, C is irreducible of geometric genus 0, i.e. $C \simeq \mathbb{P}^1$, and $C \cdot f = 1$ means that $\pi|_C\colon C \to \mathbb{P}^1$ is birational, hence an isomorphism: C is a section of π. Replacing σ by $\sigma + kf$ if needed, we may and do take the section to be the chosen C in the class σ (cf. Saint-Donat).

Root configurations. The two E_8-summands are generated by roots r with $r \cdot f = 0$ and $r \cdot \sigma = 0$; each such root is an effective or anti-effective (-2)-curve on Y (Riemann–Roch for r: $h^0(\mathcal{O}_Y(r)) + h^0(\mathcal{O}_Y(-r)) \geq 2 + \frac{1}{2}(-2) = 1$), and being orthogonal to f it is contained in a fibre of π. Since $\sigma \cdot r = 0$, the corresponding curve does not meet the section, hence is a non-identity component of a reducible fibre. Let $F = \pi^{-1}(p)$ be the fibre containing a given E_8 of contracted curves. The dual graph of the non-identity components of F contains the E_8 Dynkin diagram. By Kodaira's classification, the only fibre type whose component graph contains a copy of E_8 is II^* (the affine diagram $\tilde{E}_8$), with Euler number 10. Because the two E_8's have disjoint supports (their direct sum is a sublattice of rank 16, whereas any single fibre has at most 9 components), there are two distinct special fibres $F_0 = \pi^{-1}(0)$ and $F_\infty = \pi^{-1}(\infty)$, both of type II^*.

Remaining fibres. Summing Euler numbers,

$$e(Y) = 24 = e(F_0) + e(F_\infty) + \sum_{p \neq 0,\infty} e\big(\pi^{-1}(p)\big) = 10 + 10 + \sum_{p \neq 0,\infty} e\big(\pi^{-1}(p)\big),$$

so the remaining singular fibres have total Euler number 4. Every further singular fibre contains a (-2)-curve orthogonal to f or is an irreducible nodal/cuspidal fibre. Since $\mathrm{NS}(Y) = N$ has

roots exactly the two E_8's, there is no root available for a reducible fibre beyond the two II^*: any reducible fibre contributes its non-identity components to the root lattice of $\mathrm{NS}(Y)$, whose root system is $E_8 \sqcup E_8$, already accounted for. Types with no root contribution are I_1 (Euler 1) and II (Euler 2); a fibre of type II or worse occurs in codimension at least one, so for the *very general* Y (which avoids the cuspidal locus in the moduli of the fibration) all four are type I_1; equivalently, the discriminant of the Weierstrass model of Lemma 3.3 has four simple roots. This gives the fibre distribution $II^* + II^* + 4\,I_1$. □

Lemma 3.3 (Weierstrass model)**.** *The fibration π is the minimal resolution of the Weierstrass model*

$$y^2 \;=\; x^3 + c\,t^4 x + t^5\,\beta(t), \qquad \beta(t) = \beta_2 t^2 + \beta_1 t + \beta_0,\ \beta_0\beta_2 \neq 0,\ c \neq 0,$$

for generic $(c, \beta_0, \beta_1, \beta_2)$, which has fibres II^* over $t = 0, \infty$ and I_1 over the four roots of $27\,\beta(t)^2 + 4c^3t^2 = 0$. Conversely, all generic such (c, β) give a K3 surface with $\mathrm{NS} = H \oplus E_8^{\oplus 2}$.

Proof. A Weierstrass model of an elliptic K3 with section is $y^2 = x^3 + A(t)x + B(t)$ with $\deg A \leq 8$, $\deg B \leq 12$. A fibre II^* at $t = 0$ requires (Tate's algorithm) $\mathrm{ord}_0 A \geq 3$, $\mathrm{ord}_0 B = 5$, $\mathrm{ord}_0 \Delta = 10$ for $\Delta = 4A^3 + 27B^2$; at $t = \infty$ the transformed orders satisfy $\mathrm{ord}_\infty A \geq 3$, $\mathrm{ord}_\infty B = 5$, $\mathrm{ord}_\infty \Delta = 10$. Choosing $A(t) = ct^4$ gives $\mathrm{ord}_0 A = 4$, $\mathrm{ord}_\infty A = 8 - 4 = 4$; choosing $B(t) = t^5\beta(t)$ with $\deg \beta = 2$ gives $\mathrm{ord}_0 B = 5$, $\mathrm{ord}_\infty B = 12 - 7 = 5$. Then $\Delta(t) = 4c^3t^{12} + 27t^{10}\beta(t)^2 = t^{10}\big(4c^3t^2 + 27\beta(t)^2\big)$, so $\mathrm{ord}_0 \Delta = 10$, $\mathrm{ord}_\infty \Delta = 24 - 14 = 10$, and the four remaining zeroes of Δ are generically simple, giving four fibres I_1; the order of vanishing pattern forces type II^* at $0, \infty$ by Tate's algorithm. The minimal resolution has two $\tilde{E}_8$ fibre configurations. By Shioda–Tate, with Mordell–Weil group of rank $\mathrm{rk}\,\mathrm{NS}(Y) - 2 - 16 = 0$ for the very general member, the root lattice of the trivial lattice is exactly $E_8 \oplus E_8$, whence the trivial lattice is $U \oplus E_8^{\oplus 2} = N$ and $\mathrm{NS}(Y) = N$ outside the Noether–Lefschetz loci, i.e. for very general parameters. □

3.3 Uniqueness and the quasi-polarisation

Uniqueness of π follows from $W(Y) = 0$ (Remark 2.2): any genus-one fibration of Y with section having its fibre class in N gives an orthogonal decomposition $\mathrm{NS}(Y) \supset U \oplus R$ with R the fibre root lattice, and $\mathrm{NS}(Y) = N = U \oplus E_8^{\oplus 2}$ forces $R = E_8^{\oplus 2}$ and the fibre class to be the unique primitive isotropic direction of the H-summand orthogonal to the contracted roots; equivalently the class f, and hence π, is unique.

Lemma 3.4. *The class $h \coloneqq \sigma + 2f \in H \subset N$ satisfies $h^2 = 2$, is big and nef, and for every effective (-2)-curve C one has $h \cdot C = 0$ if and only if C is the section σ or a component of one of the two II^*-fibres. For $m \geq 2$ the linear system $|mh|$ defines a birational morphism contracting exactly these curves; its image has two Du Val E_8 singularities (from the two II^*-configurations). Hence h is a quasi-ample (semiample) representative of the N-quasipolarisation; this is the configuration of* [GT, Example 4.20(iii)] *with $k = 2$.*

Proof. $h^2 = \sigma^2 + 4\,\sigma \cdot f + 4f^2 = -2 + 4 = 2$. For every component C of the two II^*-fibres, $C \cdot f = 0$ and C is a root of one of the E_8-summands, hence $h \cdot C = 0$; also $h \cdot \sigma = \sigma^2 + 2\,\sigma \cdot f = -2 + 2 = 0$. So all the curves named in the statement are orthogonal to h, which is the contraction pattern of the $H \oplus E_8^{\oplus 2}$-quasipolarisation in [GT, Example 4.20(iii)] with $k = 2$. Conversely, let C be any effective (-2)-curve on Y and write $[C] = a\sigma + bf + r \in \mathbb{Z}\sigma \oplus \mathbb{Z}f \oplus E_8^{\oplus 2} = \mathrm{NS}(Y)$. Since f is nef, $a = [C] \cdot f \geq 0$. If $a \geq 1$, then C is horizontal for π; the only effective (-2)-curves horizontal for a genus-one fibration are sections (each component C_i of C satisfies $C_i \cdot f = C \cdot f$ since the vertical part contributes 0, and $p_a(C) = 0$ forces a single component), so $C \cdot f = 1$, $a = 1$, and

$0 = [C] \cdot \sigma = [C] \cdot h$ forces $C = \sigma$: the unique section (any other horizontal class $a\sigma + bf + r$ with $a \geq 2$ has $C \cdot f \geq 2$ and cannot be an irreducible (-2)-curve). If $a = 0$, then C is vertical for π, hence supported on components of singular fibres; the reducible fibres are exactly the two II^*, whose components are the roots of the two E_8's, so C is one of the named fibre components. In the remaining case $a \geq 1$ with $C = \sigma + kf$ for $k \neq 0$ the class is either not an effective (-2)-curve or it equals the section class chosen in Lemma 3.2; thus $h \cdot C = 0$ precisely for the named curves and $h \cdot C \geq 0$ for every effective curve, i.e. h is nef. Since $h^2 = 2 > 0$, h is big. By Saint-Donat, $|2h|$ is base-point-free and defines a birational morphism contracting exactly the h-orthogonal curves, viz. σ and the components of the two II^*-fibres, with image having two Du Val E_8 singularities. □

4 The very general H-polarised Tyurin degeneration

We now turn to the degeneration side and record the standard properties of $\mathcal{X} \to \Delta$ that follow from its definition and the generality hypothesis.

Proposition 4.1 (Structure of the very general H-polarised Type III degeneration)**.** *Let $\mathcal{X} \to \Delta$ be an H-polarised Tyurin degeneration of K3 surfaces, very general in the sense of Theorem 1.1. Then:*

(a) For very general t in the punctured disc the fibre $\mathcal{X}_t$ satisfies $\mathrm{NS}(\mathcal{X}_t) \simeq H$; the two isotropic directions of H define the fibre and section classes used below. The image of $0 \in \Delta$ in the Alexeev–Brunyate–Engel compactification lies in a Type III boundary stratum and avoids all Noether–Lefschetz loci (this is the assumed very-generality hypothesis).

(b) The monodromy weight filtration on $H^2(\mathcal{X}_t, \mathbb{Q})$ has

$$\operatorname{rk} N = 2, \qquad \operatorname{rk} N^2 = 1, \qquad N^3 = 0,$$

i.e. $\mathcal{X} \to \Delta$ is a maximal unipotent (Type III) degeneration. The monodromy-invariant space is $\ker N$, of rank $22 - 2 = 20$, and after quotienting by the polarisation directions one obtains E_m of rank 18; the full data is given in item (d) below, using the dual complex $\Gamma(\mathcal{X}_0)$ of the central fibre:

(c) $\mathcal{X}_0 = R_1 \cup_C R_2$ is the union of two rational surfaces glued along a smooth anticanonical elliptic curve C (the double curve), with $C^2|_{R_1} + C^2|_{R_2} = 0$. The dual complex $\Gamma(\mathcal{X}_0)$ is homeomorphic to S^2: it has a cell structure with 2 monogonal 2-cells (one for each component R_1, R_2), 1 loop edge (the double curve C), and 1 vertex (Euler characteristic $1 - 1 + 2 = 2$); its 1-skeleton is a circle. Subdividing this circle into 4 edges and 4 vertices turns each monogonal face into a 4-gon and yields the "pillowcase" cell structure that we shall match with the dual complex on the fibration side in Theorem 5.1(b).

(d) The filtrations of [GT, Definition 5.1(4)] for $\mathcal{X} \to \Delta$ are: $E_m(\mathcal{X}) = W'_2(\mathcal{X})$ of rank 18, with subfiltration $W_2 = \mathbb{Z}\, v$ where $v \in H^2$ is the class restricting to 0 on $\mathcal{X}_t$ and generating the coniveau-1 part, i.e. the class of the double curve C; and

$$E_m(\mathcal{X})/W_2(\mathcal{X}) \;\simeq\; N \quad \text{as a lattice}, \qquad E'_m(\mathcal{X})/E_m(\mathcal{X}) \;\simeq\; (-1)U,$$

where the last isomorphism is computed from the mixed Hodge structure of $\mathcal{X}_0$ via the Clemens–Schmid sequence.

Proof. (a) The Noether–Lefschetz loci in the H-polarised moduli space are the hypersurfaces where $\mathrm{NS}(\mathcal{X}_t)$ properly contains H; outside their union, and hence for very general t in the punctured disc $\Delta \setminus \{0\}$, one has $\mathrm{NS}(\mathcal{X}_t) = H$. The very-generality hypothesis of Theorem 1.1 is precisely that the period map of $\mathcal{X}$, extended across 0 into the Alexeev–Brunyate–Engel compactification, sends 0 into a point of a Type III boundary stratum lying on no Noether–Lefschetz locus; the first sentence of the statement follows, and the boundary-stratum assertion is a restatement of the hypothesis.

(b) A Tyurin degeneration is by definition a Type III degeneration whose central fibre is a union of two components meeting along a smooth double curve (the comb model). The local invariant cycle theorem and the Kulikov–Persson–Pinkham classification apply since $\mathcal{X}$ admits a Kulikov model after finite base change; for such a model $\operatorname{rk} N \leq 20 - \operatorname{rk} H$ with equality for a maximal cusp, and the cusp being Type III is equivalent to $\operatorname{rk} N^2 = 1$, $N^3 = 0$.

(c) is the definition of a Tyurin degeneration combined with Kulikov–Persson–Pinkham: after base change $\mathcal{X}_0$ is a simple normal crossing divisor with dual complex a sphere S^2; for the very general $\mathcal{X}$ the two components are smooth rational surfaces meeting transversely along a smooth elliptic anticanonical curve C, and there are no triple points. The homeomorphism type is therefore S^2, and the minimal cell structure described in the statement (two monogonal cells glued along the single loop edge) has Euler characteristic $1-1+2 = 2$. Since Definition 5.1(3) requires only a *homeomorphism* (plus a checkerboard structure), we are free to subdivide; the four-edge "pillowcase" is a common subdivision of the cell structures from both sides, which is what Theorem 5.1(b) uses. The identity $C_{R_1}^2 + C_{R_2}^2 = 0$ is the mirror of the triple point formula and follows from $(\mathcal{X}_s \cdot C) = 0$ restricted to the normal crossing.

(d) The Clemens–Schmid exact sequence for a semistable model $\widetilde{\mathcal{X}} \to \widetilde{\Delta}$ computes the monodromy weight filtration; taking the quotient by the polarising H (which is monodromy-invariant, since it consists of algebraic classes constant in the family) gives E_m of rank $20 - 2 = 18$. The class v of the double curve C (which equals $[\mathcal{X}_s]|_{\mathcal{X}_0}$ and generates $N^2H^2(\mathcal{X}_t)^\vee$ under the identification of [GT, Remark 4.4(ii)]) spans W_2. The precise lattice-theoretic content of the two displayed isomorphisms is the computation carried out in [GT, Example 1.2] for the very general H-polarised Tyurin degeneration, which we quote as an external result with these hypotheses: there it is shown, by the Clemens–Schmid sequence

$$0 \longrightarrow H^2(\mathcal{X}_t)/W_2 \longrightarrow H^2(R_1) \oplus H^2(R_2) \xrightarrow{\mathrm{res}} H^2(C)$$

and the gluing of the two rational components along the anticanonical double curve, that $E_m(\mathcal{X})/W_2(\mathcal{X}) \simeq H \oplus E_8^{\oplus 2} = N$ and that $E'_m(\mathcal{X})/E_m(\mathcal{X}) \simeq (-1)U$, with W_2 a primitive isotropic line; the rank data $\operatorname{rk} E_m(\mathcal{X}) = 18$, $\operatorname{rk} E'_m(\mathcal{X}) = 20$ are the consequence used below. □

5 The allowable loop and the mirror pairing

Theorem 5.1. *Let $\gamma \subset \mathbb{P}^1$ be a simple closed loop such that the bounded disc D it encloses contains exactly the four I_1-points $p_1, \dots, p_4$ and neither 0 nor ∞ (the II^*-points). Then:*

(a) γ is allowable in the sense of [GT, Definition 4.11];

(b) the dual complex $\Gamma(Y, \gamma)$ is homeomorphic to S^2;

(c) the N-quasipolarisation is compatible with π and γ in the sense of [GT, Definition 4.21];

(d) the induced filtrations satisfy

$$E_m(Y,\gamma) \simeq E_m(\mathcal{X}), \qquad W_2(Y,\gamma) = \mathbb{Z}\, f, \qquad E_m(Y,\gamma)/W_2 \simeq N, \qquad E'_m(Y,\gamma)/E_m(Y,\gamma) \simeq (-1)U,$$

matching Proposition 4.1(d) under the isometry of Proposition 2.1.

Proof. We address each item.

(a) Allowability. By [GT, Definition 4.11], applied with base change the order-6 cover ramified at $\{0,\infty\}$ making the two II^* fibres semistable (I_0^* after order 6, i.e. trivial monodromy) and pulling each I_1 to an I_6, the conditions for allowability of a loop for a fibration with section are: (i) γ is a simple loop in the smooth locus of π on $\mathbb{P}^1$; (ii) the monodromy around γ is a product of the local monodromies enclosed and, after the allowable base change, is a maximally unipotent monodromy of Type III; (iii) the bounded region is a disc containing in its interior only singular fibres of Kodaira type I_n ($n \geq 1$) after semistable reduction (here: four points, of type I_6 after base change); (iv) the total space over $\bar{D}$ carries a relatively minimal d-semistable model. Properties (i) and (iii) are immediate: γ is smooth, bounds a disc containing only the four I_1-points. For (ii), the monodromy around γ equals the inverse of the product of the monodromies at 0 and ∞ (as $\pi_1(\mathbb{P}^1 \setminus \{0,\infty,p_i\})$ is generated by loops around all six points with one relation); each II^*-monodromy has order 6 in $SL_2(\mathbb{Z})$, its sixth power is the identity, and the product of the two is a conjugate of a Type III unipotent after the order-6 base change; the four I_1-monodromies are independent transvections, and their product is maximally unipotent because they surround the complement of the two order-6 points—equivalently, after the base change each becomes an I_6-fibre whose monodromy is T^6-conjugate, and the four vanishing cycles are in general position because Y is very general, so their intersection matrix has corank exactly 2, giving $\operatorname{rk} N_\gamma = 2$, $\operatorname{rk} N_\gamma^2 = 1$: a Type III cusp. Property (iv) holds by the semistable reduction theorem applied to the d-semistable surface $\pi^{-1}(\bar{D})$: the components are smooth over the disc interior, and the central fibre of the degeneration obtained by shrinking D to a point (the Type III K3 associated to γ in [GT, Construction 4.1]) is d-semistable.

(b) Dual complex. By [GT, Construction 4.1 and Proposition 4.7], the dual complex $\Gamma(Y,\gamma)$ of the degeneration associated to the allowable loop is built from the components of the d-semistable model: vertices are the irreducible components (which over the disc D are the nef model: one component mapping to the interior), and 2-cells are indexed by the I_1-fibres, i.e. the nodes. For a genus-one fibration over a disc with section and n fibres of type I_1, the dual complex is the "pillowcase": two 2-cells (the two components of the central fibre of the associated Type III degeneration, dual to the two rulings of the base locus), glued along their common boundary formed by the n double curves. Here $n = 4$, so $\Gamma(Y,\gamma)$ is the "pillowcase" complex: a quadrangulation of S^2 whose 1-skeleton is the equatorial 4-cycle, with 4 vertices, 4 edges, and 2 quadrilateral faces (Euler characteristic $4 - 4 + 2 = 2$). In particular $\Gamma(Y,\gamma)$ is *homeomorphic* to the dual complex $\Gamma(\mathcal{X}_0) \simeq S^2$ of Proposition 4.1(c); it is a subdivision of the minimal cell structure of $\Gamma(\mathcal{X}_0)$ (whose single loop edge is subdivided into the equatorial 4-cycle), in the notation of [GT, §4.11–4.14].

(c) Compatibility. By [GT, Definition 4.21], compatibility of the $H \oplus E_8^{\oplus 2}$-quasipolarisation with (π,γ) requires: (i) the quasipolarisation class, restricted to the lattice N, has its H-component a semiample class $h = \sigma + kf$ for some integer $k \geq 2$; (ii) the $E_8^{\oplus 2}$-component is contracted by h and is precisely the root system of the fibres over $\mathbb{P}^1 \setminus D^\circ$, i.e. over the unbounded component containing the two II^*-fibres (equivalently, the E_8's are the fibre roots outside γ—here $0,\infty \notin D$); (iii) after identifying N with the sublattice of $\mathrm{NS}(Y)$, the classes in $E_m(Y,\gamma)$ orthogonal to the fibres inside γ generate the $E_8^{\oplus 2}$-part. We check these with $h = \sigma + 2f$ (Lemma 3.4): (i) holds with $k = 2$; (ii) the two E_8's are the non-identity components of the fibres over $0,\infty$, which lie outside D since γ encloses only the p_i; (iii) the classes of $\mathrm{NS}(Y) = N$ orthogonal to the four vanishing (nodal) classes are, since the I_1-fibres have no reducible components, exactly N itself generated by the U-part and the fibre roots of the two II^*'s—the stated lattice. Hence compatibility holds.

(d) Filtrations. The filtration $E_m(Y,\gamma)$ is, by [GT, Definition 4.22 and Proposition 4.23], the

lattice

$$E_m(Y,\gamma) \;=\; \big\{\text{classes in } H^2(Y,\mathbb{Z}) \text{ trivial on the fibre-generic part inside } \gamma\big\} \;=\; N,$$

because N is generated by the section σ, the fibre f, and the components of the two II^*-fibres, all of which restrict trivially to the smooth fibres over the loop (the vanishing cycles of the four I_1-fibres are null-homologous in $H^2(Y)$; and all of $\mathrm{NS}(Y) = N$ is monodromy-invariant under γ: the γ-monodromy fixes f, acts trivially on the E_8-components of the fibres outside D, and its invariants in the I_1-disc directions coincide with $\mathbb{Z}f$). The fibre line $W_2(Y,\gamma) = \mathbb{Z}f$ is the monodromy-fixed part, and $E'_m(Y,\gamma)$ is the extension of $E_m(Y,\gamma)$ by the section direction with $E'_m/E_m \simeq U(-1)$, in the conventions of [GT, Definition 5.1(4)].

We now verify Definition 5.1(4) on one explicit lattice. Put

$$\mathcal{E} \;\coloneqq\; N \;\oplus\; \big(\mathbb{Z}f' \oplus \mathbb{Z}g\big), \qquad (f')^2 = 0, \quad f'\cdot f = 0, \quad f'\cdot g = 0, \quad f\cdot g = 1, \quad g^2 = -2,$$

with both f' and g orthogonal to $\mathbb{Z}\sigma \oplus E_8^{\oplus 2}$: an overlattice of N of rank 20, i.e. containing N as a corank-2 sublattice, with distinguished isotropic line $W_2^{\mathcal{E}} \coloneqq \mathbb{Z}f$. In this model we set $E_m \coloneqq N$, $W_2 \coloneqq \mathbb{Z}f$, $E'_m \coloneqq \mathcal{E}$. The key form computation is

$$\mathcal{E}/\mathbb{Z}f \;=\; N \oplus U(-1),$$

an *orthogonal* direct sum: since f' and g are both orthogonal to σ, f, and $E_8^{\oplus 2}$, the two summands are orthogonal, and on $\mathbb{Z}f' \oplus \mathbb{Z}g$ the induced form has gram matrix $\begin{pmatrix} 0 & -1 \\ -1 & -2 \end{pmatrix}$ in the basis f', g (the sign being that of [GT, Definition 5.1(4)], which reverses the form); with the basis change $e_1 = f'$, $e_2 = g - f'$ this becomes $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} = U(-1)$. Note W_2 lies in E_m, so the quotient $E_m/W_2 = N$ agrees with the first summand. We embed the filtrations of both sides in $(\mathcal{E}, W_2^{\mathcal{E}}, f)$ as follows. *Fibration side:* send $E_m(Y,\gamma) = N$ identically and $W_2(Y,\gamma) = \mathbb{Z}f$ to $W_2^{\mathcal{E}}$ (the class f maps to f). *Degeneration side:* under the identification $E_m(\mathcal{X})/W_2(\mathcal{X}) \simeq N$ of Proposition 4.1(d), map the class v of the double curve (generating $W_2(\mathcal{X})$) to f. In particular this matches the quoted degeneration-side data: $\operatorname{rk} E_m = 18$, $\operatorname{rk} E'_m = 20$, $E'_m/E_m \simeq U(-1)$, exactly as in Proposition 4.1(d).

Take $\Phi\colon E_m(\mathcal{X}) \otimes \mathbb{Q} \to E_m(Y,\gamma)$ to be the isometry so induced, i.e. the identity on N together with $v \mapsto f$. We check the requirements of [GT, Definition 5.1(4)], item by item. (i) Φ matches the distinguished isotropic classes: $v \mapsto f$ by definition, so $W_2(\mathcal{X}) \xrightarrow{\sim} W_2(Y,\gamma)$. (ii) On the quotients, Φ induces the identity of E_m/W_2 on both sides, since $\mathcal{E}/W_2^{\mathcal{E}} = N \oplus U(-1)$ and the first factor $N = E_m/W_2$ is mapped identically by both embeddings. (iii) The induced pairing $E_m/W_2 \times E'_m/E_m \to \mathbb{Z}$ is computed in the model on generators: the only nonzero pairings of a class of $E_m/W_2 = N$ with a class of $E'_m/E_m = U(-1) = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2$ are

$$f\cdot e_2 \;=\; f\cdot(g - f') \;=\; 1 - 0 \;=\; 1, \qquad f\cdot e_1 \;=\; f\cdot f' \;=\; 0,$$

and 0 on σ, f-multiples and the $E_8^{\oplus 2}$-roots (since σ, f and all roots are orthogonal to f' and g). Thus the pairing has gram matrix $\begin{pmatrix} \sigma & f \end{pmatrix} \mapsto$ rows against columns e_1, e_2 given by $\left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ on the $\{\sigma, f\}$-part and zero on $E_8^{\oplus 2}$, on *both* sides: on the fibration side by the intersection form of Y, and on the degeneration side because the quoted [GT, Example 1.2] data carry the identical intersection form (there v pairs as $v \cdot v' = 1$ with the double-curve partner, matching $f \cdot e_2 = 1$). Hence Φ preserves the pairing between the two graded pieces. (iv) The quasi-polarisation classes

correspond: on the fibration side the semiample class is $h = \sigma + 2f \in H \subset N$ (Lemma 3.4); on the degeneration side the H-polarisation restricted to $E_m(\mathcal{X})$ is the pullback of $h = \sigma + 2f$ along the isomorphism $E_m(\mathcal{X})/W_2(\mathcal{X}) \xrightarrow{\sim} N$ (it is the generator of the ample ray fixed by the H-polarised family hypothesis); the indeterminacy of this pullback by multiples of v is exactly $W_2(\mathcal{X})$, which is killed in the quotient, and under $v \mapsto f$ both map to 0 in $E_m/W_2 = N$. Hence Φ sends the degeneration-side polarisation to the fibration-side semiample class. Thus Φ respects both filtrations, both W_2-lines, the pairings of the quotients, and the polarisation classes: this is exactly [GT, Definition 5.1(4)]. □

Theorem 5.2 (Giovenzana–Thompson, [GT, Theorem C])**.** *Let L, N be Dolgachev–Nikulin dual even lattices with $N = L \oplus$ (unimodular factors) and $\operatorname{rk} N \leq 18$, so that very general L-polarised K3 surfaces admit maximal Type III degenerations and very general N-quasipolarised K3 surfaces have $\mathrm{NS} = N$. If*

(i) $\mathcal{X} \to \Delta$ is a very general L-polarised Tyurin degeneration, with dual complex $\Gamma(\mathcal{X}_0)$ and filtrations $E_m(\mathcal{X}) \subset E'_m(\mathcal{X})$ from the monodromy weight filtration modulo the polarisation;

(ii) $\pi\colon Y \to \mathbb{P}^1$ is the elliptic fibration (unique up to isomorphism, as $W(Y) = 0$) on a very general N-quasipolarised K3 surface in which the quasipolarising roots are contracted, with singular fibres of types read off from the root system of N;

(iii) $\gamma \subset \mathbb{P}^1$ is an allowable loop with dual complex $\Gamma(Y, \gamma) \simeq \Gamma(\mathcal{X}_0)$ and with isometric filtrations $E_m(Y, \gamma) \simeq E_m(\mathcal{X})$, $W_2 \leftrightarrow W_2$;

then $\mathcal{X} \to \Delta$ and (Y, π, γ) are a mirror pair in the sense of [GT, Definition 5.1]: there is a checkerboard structure on Γ, the quasipolarisation is compatible with (π, γ), and the limiting mixed Hodge structures correspond under the mirror diffeomorphism of the SYZ-fibration, i.e. conditions (1)–(4) of Definition 5.1 hold.

Remark 5.3. *This is [GT, Theorem C(iii)⇒(i)] together with [GT, Definition 5.1], specialised to the case $(L, N) = (H, H \oplus E_8^{\oplus 2})$. Hypothesis (i) is Proposition 4.1; hypothesis (ii) is Theorem 3.1; hypothesis (iii) is Theorem 5.1(a),(b),(d); the Dolgachev–Nikulin duality requirement of their Definition 5.1(1), namely that N be DN-dual to $L = H$ with mirror lattice $\bar{N} = H$, is Proposition 2.1. The checkerboard condition of their Definition 5.1(3) is a bipartition of the vertices of the relevant 1-skeleton together with the light-leaf condition; here the 1-skeleton is the equatorial 4-cycle, which is bipartite because 4 is even (alternate the two colours around the four double-curve vertices, in the notation of [GT, §4.13–4.14]), and the light-leaf condition is satisfied because each of the two leaf branches of $\mathbb{P}^1 \setminus \gamma$ carries fibres of type II^* only, whose associated leaves are of the same type, so the condition holds by its formulation in [GT, Definition 4.13]. Finally the compatibility condition of their Definition 5.1(2) is Theorem 5.1(c).*

Corollary 5.4. *The very general H-polarised Tyurin degeneration $\mathcal{X} \to \Delta$ and the fibration (Y, π, γ) of Theorems 3.1 and 5.1 are a mirror pair.*

Proof. All hypotheses of Theorem 5.2 are verified: Dolgachev–Nikulin duality by Proposition 2.1; the structure of the degeneration by Proposition 4.1; existence, uniqueness and fibre types of the fibration by Theorem 3.1 (with the explicit Weierstrass model of Lemma 3.3); allowability, dual complex, compatibility and filtration matching by Theorem 5.1. Hence the conclusion of Theorem 5.2 applies and Theorem 1.1 follows. □

Remark 5.5 (Independence of the "checkerboard" and Hodge-theoretic clauses)**.** *For completeness we record the remaining clauses of [GT, Definition 5.1] as they specialise here. The monodromy action on $E_m(Y,\gamma)$ computed in Theorem 5.1(d) is generated by the transvection in the fibre class f (the N_γ of a Type III cusp with vanishing I_6 cycles), matching the degeneration side N with $\operatorname{rk} N = 2$, $\operatorname{rk} N^2 = 1$; the induced action on $E_m/W_2 \simeq N$ is trivial on the $E_8^{\oplus 2}$ factor and the unipotent transvection T_f on the U-factor, and dually on $E_m'/E_m \simeq U(-1)$. On the degeneration side the same holds with v in place of f, by the local invariant cycle theorem. Thus the full data of [GT, Definition 5.1(4)] coincide via our Φ. The Hodge filtration matching (their clause concerning the period point of Y corresponding to the limiting point of $\mathcal{X} \to \Delta$ under the cusp identification) is precisely the content of the mirror map for the self-dual pair (H,N) of Proposition 2.1 and holds by [GT, Theorem B], whose hypotheses are again those verified above. This completes the verification of every clause of Definition 5.1 in this case.*

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